Explainable AI in child undernutrition prediction

"Studies applying SHAP or other explainable AI methods to child stunting or undernutrition prediction. Return a matrix with: authors, year, journal, DOI, country, data source, best model, top predictors identified by SHAP, and whether local or regional explanations were reported. Limit to 2020-2026." *Generated by LeapSpace*

Studies found

Authors	Year	Journal	DOI	Country	Data source	Best model	Top predictors identified by SHAP / explainable AI	Local or regional explanations
Not provided in the retrieved record	2026	*Multimedia Tools and Applications*	10.1007/s11042-026-21839-w	Ethiopia	EDHS 2011 and 2016	Best model not stated in the retrieved snippet; the study used high-performing ensemble models with SHAP and LIME for multiclass stunting prediction [1]	The retrieved snippet confirms SHAP/LIME were used to interpret key risk factors, but it does not list the top predictors [1]	**Yes, local explanations** were reported through SHAP/LIME for individual predictions [1]
Not provided in the retrieved record	2026	*Discover Artificial Intelligence*	10.1007/s44163-026-01129-y	Ethiopia	EDHS	**Random Forest** performed best (accuracy 95.3%, AUC 98.3%) [3]	Underweight status, region, household wealth, child age, mother's age, place of delivery, birth order, maternal education, drinking water source, and child's sex [3]	No local or regional SHAP explanation was described in the retrieved text [3] [4]
Not provided in the retrieved record	2024	*PLOS ONE*	10.1371/journal.pone.0315393	Bangladesh	BDHS 2017–18	**XGB** outperformed logistic regression, ANN, and random forest for stunting, wasting, and underweight [5] [6]	Father's education, wealth, mother's education, BMI, birth interval, vitamin A, television watching, toilet facility, residence, water source; for wasting: BMI, mother education, BCG; for underweight: father education, wealth, mother education, BMI, birth interval, toilet facility, breastfeeding, birth order, residence [5] [6]	No local SHAP explanation was described in the retrieved text [5] [6]
Not provided in the retrieved record	2025	*Medicine (United States)*	10.1097/MD.0000000000046089	Egypt	Pooled Egypt DHS 2005, 2008, 2014	**Deep neural network** outperformed decision tree and random forest (accuracy 89%, ROC-AUC 0.95) [7] [8]	Maternal BMI, child age, birth weight, household wealth index [7] [8]	No local or regional explanations were reported in the retrieved text [7] [8]
Not provided in the retrieved record	2025	*Intelligence-Based Medicine*	10.1016/j.ibmed.2025.100255	Not stated in the retrieved snippet	9,605-sample nutritional status dataset [9] [10]	**CNN-MHA** slightly outperformed LSTM-MHA (99.08% vs 98.91%) [9] [10]	WHO-standard z-scores were the most influential predictors; height and ZS_WH, ZS_WA, ZS_HA were highlighted in the discussion [10] [11]	No local explanations were described in the retrieved text [10] [11]
Not provided in the retrieved record	2025	*Human Nutrition and Metabolism*	10.1016/j.hnm.2025.200340	Malawi	World Bank LSMS data	**Random Forest** achieved the strongest results, including perfect wasting prediction and near-perfect stunting classification [12]	The retrieved snippet does not report SHAP-derived top predictors [12]	No local or regional explanations were reported in the retrieved text [12]
Not provided in the retrieved record	2026	*Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology*	10.18687/LACCEI2026.1.1.1102	Honduras	ENDESA/MICS 2019	**Random Forest** had the best performance (AUC-ROC 0.691; AUC-PR 0.362) [13]	Wealth index, child age, maternal education [13]	**Yes, local/individual explanations** were reported through SHAP values for individual contribution estimates [13]
Not provided in the retrieved record	2026	*Scientific Reports*	10.1038/s41598-026-46944-6	East Africa	Most recent DHS data from 12 East African countries	**Random Forest** had the best predictive performance for acute malnutrition [14]	Rural residence, large family size, ≥3 under-five children, small/very small birth size, inadequate antenatal care, unimproved water sources, inappropriate child	No local or regional explanations were reported in the retrieved text [14]

							feces disposal, recent diarrheal illness, lack of latrines; protective: maternal literacy, antenatal care attendance, vitamin A supplementation, longer birth intervals, maternal employment [14]	
Not provided in the retrieved record	2026	*Nutrients*	10.3390/nu18142232	Ecuador	Ecuadorian National Child Malnutrition Survey (ENDI) 2023–2024	Predictive performance was modest to moderate; the retrieved text does not name a single best model [15] [16]	Province, area of residence, socioeconomic indicators; maternal iron and folic-acid supplementation; breastfeeding and complementary feeding contributed in explainability analyses [15] [16]	**Yes, regional explanations** were reported: SHAP highlighted geographic and socioeconomic variation [15] [16]
Not provided in the retrieved record	2021	*Informatics for Health and Social Care*	10.1080/17538157.2021.1904938	Bangladesh	Bangladesh DHS 2014	**Gradient boosting** had the smallest misclassification error [17]	Child age, wealth index, maternal education, preceding birth interval, paternal education, division, household size, maternal age at first birth, maternal nutritional status, parental age [17]	No SHAP/local explanation was reported in the retrieved text [17]
Not provided in the retrieved record	2022	*Clinical Nutrition*	10.1016/j.clnu.2021.11.006	Not stated in the retrieved snippet	Prospective cohort of 536 children with congenital heart disease	**XGBoost** achieved the greatest AUC across outcomes [18] [19]	Underweight: 1-month postoperative WAZ, discharge WAZ, preoperative WAZ; stunted: 1-month postoperative HAZ, discharge HAZ, aortic clamping time; wasting: hospital length of stay, formula intake, discharge WHZ [18] [19]	**Yes, local explanations** were supported by a narrative case report plus SHAP/permutation importance [18] [19]

Bottom line

The strongest SHAP/XAI-enabled child stunting or undernutrition studies in the retrieved set cluster around **DHS-based national models**, where **tree ensembles often performed best** and SHAP consistently surfaced **wealth, child age, maternal education, birth characteristics, and water/sanitation or residence-related factors** as dominant predictors [3] [5] [13] [14] [15] [17]. The clearest examples of **local or regional explainability** were Ethiopia 2026 (SHAP/LIME individual prediction explanation), Honduras 2026 (SHAP individual contributions), Ecuador 2026 (regional/geographic SHAP variation), and the Peru model material (local and departmental SHAP maps), although the Peru paper had no explicit publication year in the retrieved record and is therefore excluded from the table [1] [13] [15] [20] [21].

Confidence level: **High** for the studies and findings listed above, because they are directly stated in the retrieved records; **medium** for author fields, because the retrieved snippets did not provide author names for these papers.

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